THE NOVA RESEARCH CURRICULUM

Biology

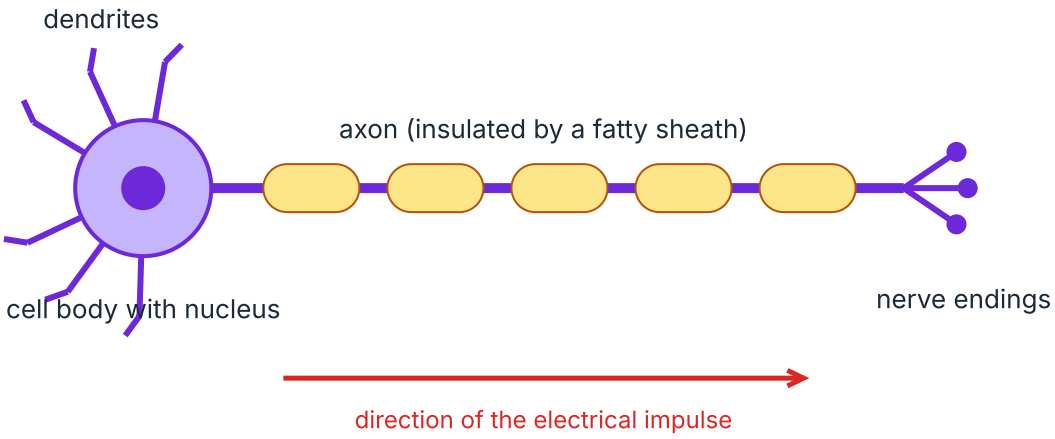
Grade 8 · Term 2

Student edition

Outcome: BI-8.2 the nervous system and reflex arcs.

Lesson notes

- **Neurons (Weeks 1–4):** nerve cells carry electrical impulses. Sensory neurons carry messages from receptors (eyes, ears, skin) to the central nervous system; motor neurons carry messages to effectors (muscles, glands); relay neurons link them.



A motor neuron.

- **Brain and spinal cord (Weeks 5–8):** the central nervous system. Cerebrum (thinking, memory), cerebellum (balance), brain stem (breathing, heartbeat). Protect the brain: helmets, seatbelts; harm from drugs and lack of sleep.
- **Reflex arc (Weeks 9–12):** a fast automatic response: stimulus → receptor → sensory neuron → relay neuron in the spinal cord → motor neuron → effector → response (pulling your hand from a hot pan). Reaction-time practical with a dropped ruler; compare left and right hands.

▶ **Try it on NovaSchool:** Reflex arc and reaction time — open **Simulations** on your dashboard.

Workbook pages

Page	Task
1	Label a neuron and the three types.
2	Brain-parts map.
3–4	Reflex-arc flow diagram; ruler-drop practical report.

Question bank

#	Outcome	Question	Your answer
1	BI-8.2	Which neuron carries messages to muscles?	sensory / relay / motor / receptor
2	BI-8.2	The central nervous system is the brain and …	heart / spinal cord / lungs / nerves in the hand
3	BI-8.2	Which part of the brain controls balance?	cerebrum / cerebellum / brain stem / spinal cord
4	BI-8.2	A reflex action is fast and automatic. True or false.	True / False
5	BI-8.2	In a reflex arc, the effector is usually a …	muscle / bone / receptor / neuron
6	BI-8.2	Put in order: motor neuron, receptor, relay neuron, sensory neuron.	_____
7	BI-8.2	Why are reflexes useful?	_____
8	BI-8.2	Wearing a helmet helps protect the brain. True or false.	True / False